Seminar 6: FIR Filters

Questions and Final Answers

1. A moving-average filter

Consider the causal three-point moving-average filter $y[n] = (1/3)(x[n] + x[n-1] + x[n-2])$

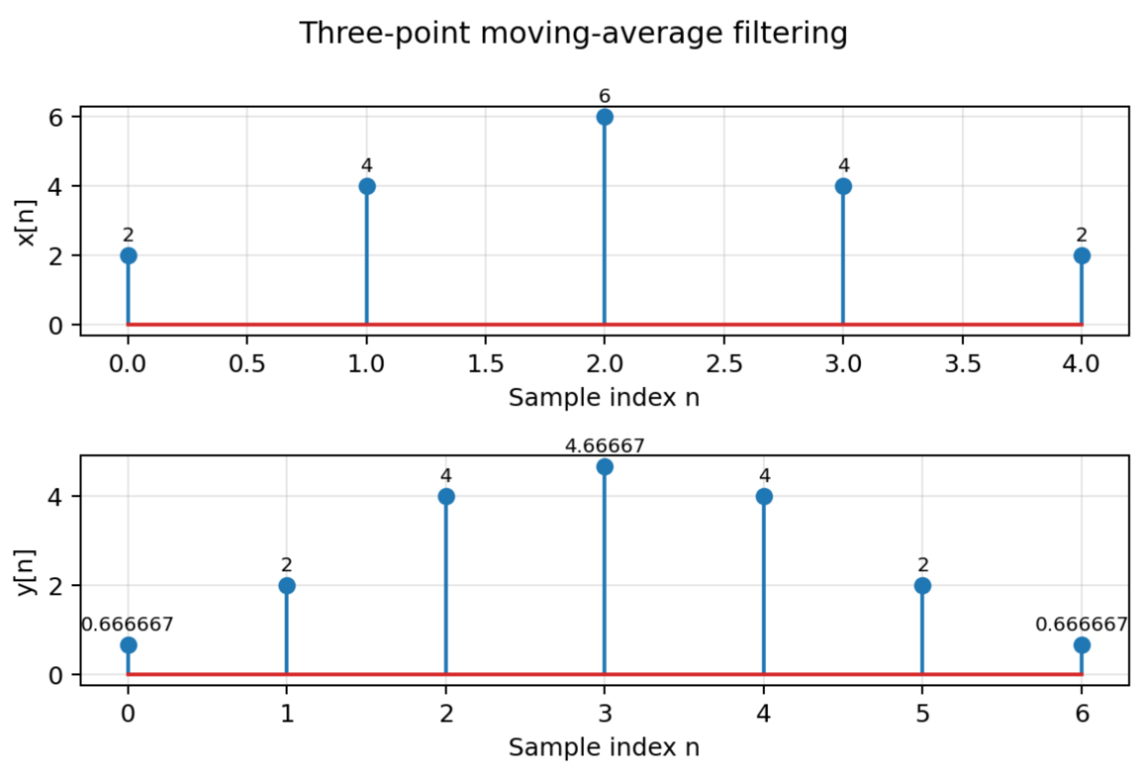
For the finite input sequence $x[n] = \{2,4,6,4,2\},\ \ 0 \le n \le 4,\ \ x[n] = 0$ otherwise

(a) Compute y[n] for all n for which the output can be nonzero.

Answer: $y[n] = \{2/3, 2, 4, 14/3, 4, 2, 2/3\},\ \ 0 \le n \le 6$

(b) Plot x[n] and y[n] as stem plots and describe the smoothing effect of the moving average.

Solution. The output has reduced sample-to-sample variation and a broader, smoother peak.



(c) Is the filter causal? Explain from the difference equation.

Solution. Yes. It uses x[n], x[n−1], and x[n−2] only.

(d) State the order and length of the filter.

Answer: $M = 2,\ \ L = M + 1 = 3$

(e) Write the impulse response h[n] of the filter.

Answer: $h[n] = (1/3)\delta[n] + (1/3)\delta[n-1] + (1/3)\delta[n-2]$

2. From coefficients to an FIR system

An FIR filter has coefficients $b[0] = 3,\ \ b[1] = -1,\ \ b[2] = 2,\ \ b[3] = 1$

(a) Write the difference equation relating y[n] to the input samples.

Answer: $y[n] = 3x[n] - x[n-1] + 2x[n-2] + x[n-3]$

(b) State the filter order M and filter length L.

Answer: $M = 3, \ \ L = 4$

(c) Write h[n] as a finite sequence and also as a sum of shifted unit impulses.

Answer: $h[n] = \{3, -1,2,1\}$

Answer: $h[n] = 3\delta[n] - \delta[n-1] + 2\delta[n-2] + \delta[n-3]$

(d) For the input x[n]={1,2,0,−1}, with x[n]=0 outside 0≤n≤3, calculate the complete output sequence.

Answer: $y[n] = \{3,5,0,2,3, -2, -1\}$

(e) Identify the three equivalent descriptions of this FIR filter represented in parts (a)–(c).

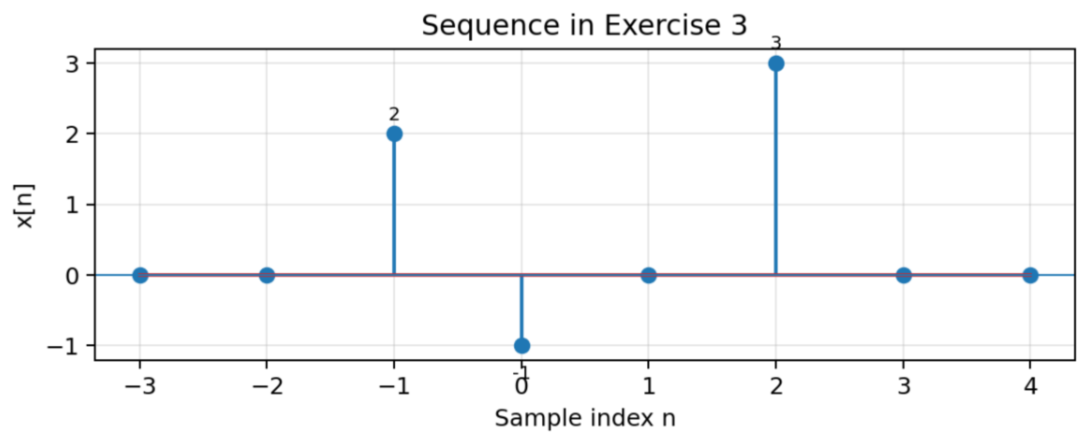
Solution. The coefficient list, the difference equation, and the impulse response.

3. Unit impulses and decomposition of a sequence

Consider the discrete-time sequence $x[n] = 2\delta[n+1] - \delta[n] + 3\delta[n-2]$

(a) Sketch x[n] for −3≤n≤4.

Solution. Nonzero samples occur at n=−1, 0, and 2 with amplitudes 2, −1, and 3, respectively.



(b) List the nonzero sample values and their indices.

Answer: $x[-1] = 2, \ \ x[0] = -1, \ \ x[2] = 3$

(c) Write the shifted sequence x[n−2] as a sum of shifted impulses.

Answer: $x[n-2] = 2\delta[n-1] - \delta[n-2] + 3\delta[n-4]$

(d) For a general sequence x[n], write the impulse-decomposition formula that represents x[n] as a sum of scaled, shifted impulses.

Answer: $x[n] = \Sigma\_(k = -\infty)\hat{}\infty \ x[k]\delta[n-k]$

(e) Explain why the impulse decomposition is useful for analysing an LTI system.

Solution. Because an LTI system's response to every shifted impulse is a shifted copy of h[n], and linearity lets the responses be summed.

4. Impulse response and delayed impulses

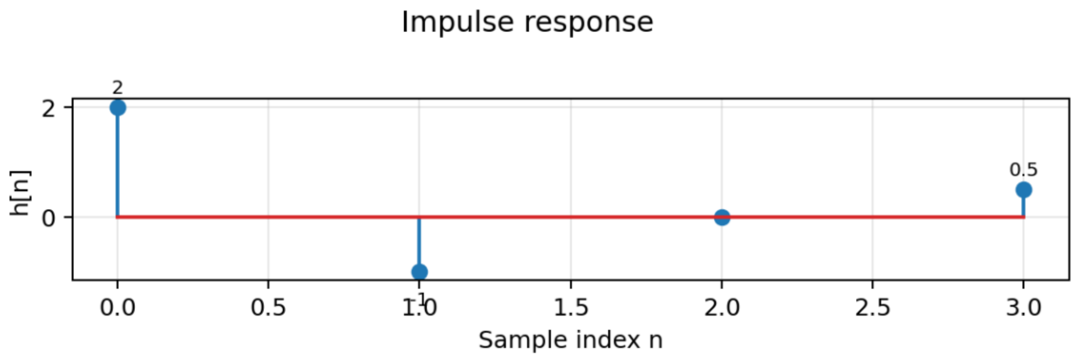
A causal FIR system is described by $y[n] = 2x[n] - x[n-1] + (1/2)x[n-3]$

(a) Find the impulse response h[n] by setting x[n]=δ[n].

Answer: $h[n] = 2\delta[n] - \delta[n-1] + (1/2)\delta[n-3]$

(b) Sketch h[n].

Solution. The nonzero samples are h[0]=2, h[1]=−1, and h[3]=1/2.



(c) Determine the output when the input is δ[n−4].

Answer: $y[n] = h[n-4] = 2\delta[n-4] - \delta[n-5] + (1/2)\delta[n-7]$

(d) Determine the output when the input is 3δ[n+2].

Answer: $y[n] = 3h[n+2] = 6\delta[n+2] - 3\delta[n+1] + (3/2)\delta[n-1]$

(e) Explain how the answers to parts (c) and (d) illustrate time invariance and linearity.

Solution. Shifts of the input cause equal shifts of h[n], and multiplying the input by 3 multiplies the output by 3.

5. Numerical convolution

Let $x[n] = \{1,2,3,2\}$, $h[n] = \{2,-1,1\}$ with both sequences beginning at n=0.

(a) Compute y[n]=x[n]*h[n] using the convolution sum.

Answer: $y[n] = \{2,3,5,3,1,2\}$

(b) Show the calculation as a sum of scaled and shifted copies of x[n].

Answer: $y[n] = 2x[n] - x[n-1] + x[n-2] = \{2,3,5,3,1,2\}$

(c) What is the length of the output sequence? Verify it from the input and impulse-response lengths.

Answer: $L\_y = L\_x + L\_h - 1 = 4 + 3 - 1 = 6$

(d) Repeat the convolution in the opposite order, h[n]*x[n], and verify that the same result is obtained.

Answer: $h[n] * x[n] = \{2,3,5,3,1,2\}$

(e) State the convolution property illustrated by part (d).

Answer: $x[n] * h[n] = h[n] * x[n] \quad \text{(commutativity)}$

6. Convolution from the impulse response

An LTI system has impulse response $h[n] = \delta[n] + 2\delta[n-1] - \delta[n-2]$

The input is $x[n] = \delta[n] + \delta[n-2]$

(a) Compute the output by applying the system separately to each shifted impulse in x[n].

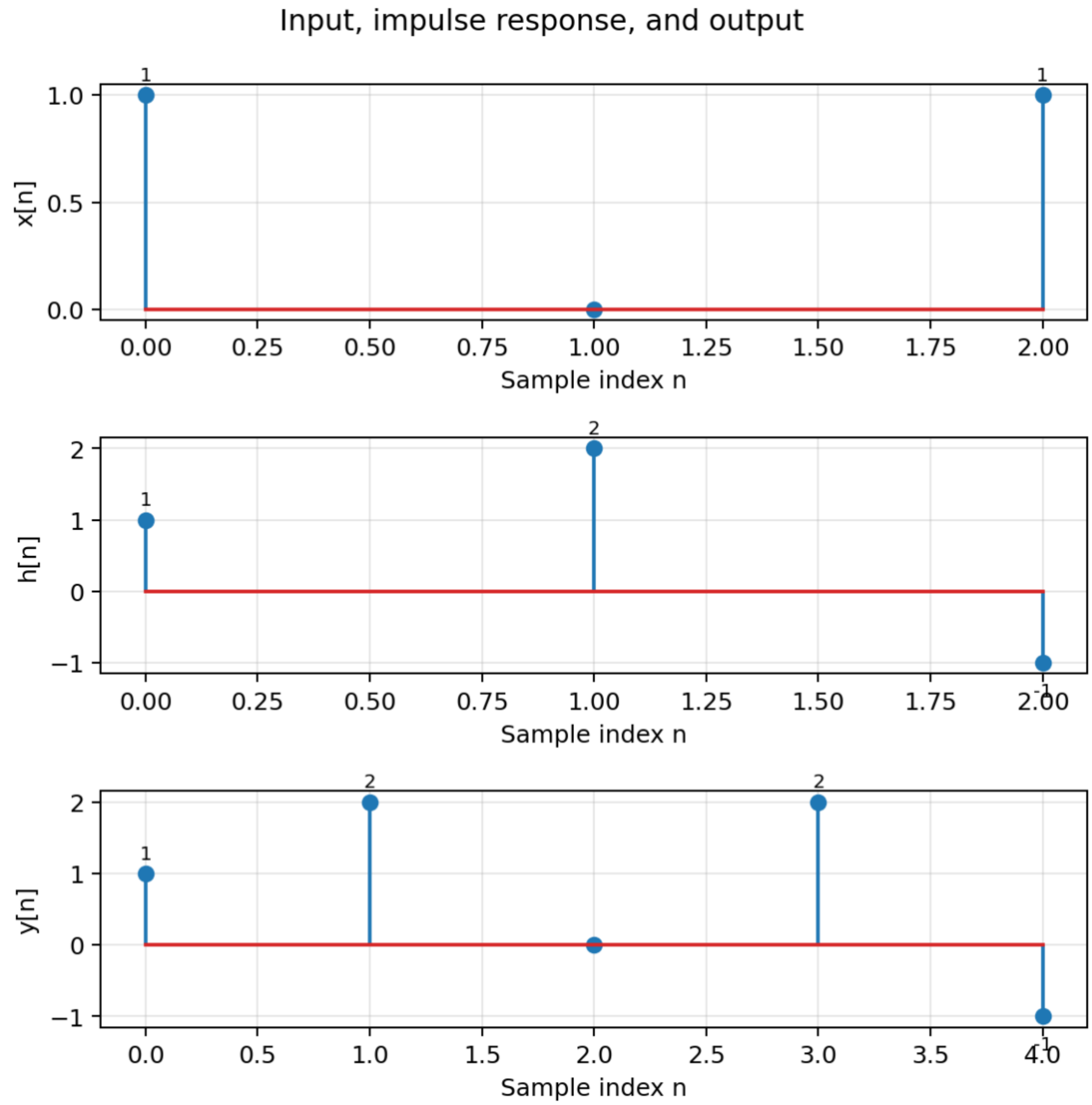
Answer: $y[n] = h[n] + h[n-2] = \delta[n] + 2\delta[n-1] + 2\delta[n-3] - \delta[n-4]$

(b) Compute the same output directly from the convolution x[n]*h[n].

Answer: $y[n] = \{1,2,0,2,-1\}$

(c) Sketch x[n], h[n], and y[n].

Solution. The stem plots are shown below.



(d) Explain why the two approaches must agree for an LTI system.

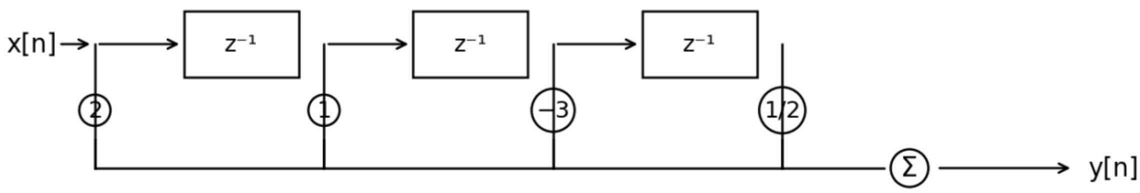
Solution. Both are the same convolution calculation expressed in two ways.

7. Direct-form FIR implementation

Consider the third-order FIR filter $y[n] = 2x[n] + x[n-1] - 3x[n-2] + (1/2)x[n-3]$

(a) Draw a direct-form block diagram using unit delays, multipliers, and adders.

Solution. A direct-form realization is shown below.



(b) How many unit-delay elements are required?

Answer: 3

(c) How many multipliers and coefficient values are required?

Solution. Four multipliers, with coefficients 2, 1, −3, and 1/2.

(d) How many additions are required to combine the four branches?

Answer: 3 additions

(e) Label the signal at the output of each delay element.

Answer: $x[n-1],\;\; x[n-2],\;\; x[n-3]$

8. Causality of FIR systems

For each of the following systems, determine whether it is causal. Give a brief reason in each case.

(i) $y[n] = x[n] + x[n-1]$

(ii) $y[n] = (1/2)x[n-1] + (1/2)x[n+1]$

(iii) $y[n] = x[n-3]$

(iv) $y[n] = x[n] + 2x[n-2] - x[n-5]$

Solution. (i) causal; (ii) noncausal; (iii) causal; (iv) causal.

(a) For every causal system, identify the largest delay used.

Solution. (i): 1 sample; (iii): 3 samples; (iv): 5 samples.

(b) For the noncausal system, identify the future sample on which the output depends.

Answer: $x[n+1]$

(c) Explain why a centred moving average is generally noncausal when implemented in real time.

Solution. Because it requires one or more future samples that have not yet arrived.

(d) Describe one situation in which a noncausal FIR operation can nevertheless be used.

Solution. Offline processing of a stored recording or data sequence.

9. Testing linearity and time invariance

Determine whether each system is linear and whether it is time invariant. Justify each answer using the definitions of linearity and time invariance.

(i) $y[n] = 3x[n] - 2x[n-1]$

(ii) $y[n] = x[n]\text{^}2$

(iii) $y[n] = n\,x[n]$

(iv) $y[n] = x[n-2] + 5$

(v) $y[n] = x[n] + x[n-4]$

Solution. (i) linear and time invariant; (ii) nonlinear but time invariant; (iii) linear but time varying; (iv) nonlinear but time invariant; (v) linear and time invariant.

(a) Which of these systems are LTI?

Solution. Systems (i) and (v).

(b) For each LTI system, write its impulse response.

Answer: $h\_i[n] = 3\delta[n] - 2\delta[n-1]$

Answer: $h\_v[n] = \delta[n] + \delta[n-4]$

(c) Which of the LTI systems are also causal?

Solution. Both (i) and (v) are causal.

10. Using linearity and time invariance without knowing the filter

An unknown LTI system produces the following input-output pair: $x_0[n] \;\rightarrow\; y_0[n]$

Without determining a difference equation or an impulse response, express the outputs for the following inputs in terms of $y_0[n]$.

(a) $2x_0[n]$.

Answer: $2y_0[n]$

(b) $x_0[n-3]$.

Answer: $y_0[n-3]$

(c) $-x_0[n+2]$.

Answer: $-y_0[n+2]$

(d) $3x_0[n]-2x_0[n-4]$.

Answer: $3y_0[n] - 2y_0[n-4]$

(e) Explain which property of an LTI system is used in each step.

Solution. Scaling and addition use linearity; shifts use time invariance.

11. The first-difference filter

Consider the FIR filter $y[n] = x[n] - x[n-1]$

(a) Write its impulse response h[n].

Answer: $h[n] = \delta[n] - \delta[n-1]$

(b) Find the output for the unit-step input u[n].

Answer: $y[n] = u[n] - u[n-1] = \delta[n]$

(c) Explain why this filter responds strongly at a sudden change in the input but gives zero for a constant sequence away from a transition.

Solution. It computes the sample-to-sample change $x[n]-x[n-1]$.

(d) For $x[n]=\{0,0,2,2,2,-1,-1\}$, beginning at n=0, calculate y[n] and identify the locations of the changes.

Answer: $y[n] = \{0,0,2,0,0,-3,0,1\},\quad 0 \le n \le 7$

Solution. The internal changes occur at n=2 (+2) and n=5 (−3); n=7 is the boundary transition from −1 back to zero.

(e) State the order and length of the filter.

Answer: $M = 1,\ \ L = 2$

12. Cascaded FIR systems

Two causal FIR systems are connected in cascade. Their impulse responses are $h_1[n] = \{1,1,1,1\},\ \ h_2[n] = \{1,1,1\}$ with both sequences beginning at n=0.

(a) Compute the equivalent impulse response $h\_eq[n]=h_1[n]*h_2[n]$.

Answer: $h\_eq[n] = \{1,2,3,3,2,1\}$

(b) State the order and length of the equivalent FIR filter.

Answer: $L = 6,\ \ M = 5$

(c) Show that interchanging the order of the two systems gives the same equivalent impulse response.

Answer: $h_2[n] * h_1[n] = \{1,2,3,3,2,1\}$

(d) Explain which property of convolution guarantees the result in part (c).

Solution. Commutativity of convolution.

(e) If a third FIR system $h_3[n]$ is added to the cascade, explain how associativity allows the equivalent response to be computed in stages.

Answer: $(h_1 * h_2) * h_3 \ = \ h_1 * (h_2 * h_3)$

13. Moving averages, smoothing, and delay

A slowly varying measurement is contaminated by rapid sample-to-sample fluctuations. Two causal moving-average filters are considered:

$$y_3[n] = (1/3)\Sigma\_(k = 0)\char`^2\ x[n - k]$$
$$y_7[n] = (1/7)\Sigma\_(k = 0)\char`^6\ x[n - k]$$

(a) State the order and length of each filter.

Solution. 3-point: M=2, L=3. 7-point: M=6, L=7.

(b) Which filter would normally give stronger smoothing? Explain in terms of the number of samples being averaged.

Solution. The 7-point average.

(c) Estimate the visible delay, in samples, introduced by each causal moving average for a slowly varying waveform.

Solution. Approximately 1 sample for the 3-point average and 3 samples for the 7-point average.

(d) Write the corresponding centred three-point average and state whether it is causal.

Answer: $y\_c[n] = (1/3)(x[n - 1] + x[n] + x[n + 1])$

Solution. It is noncausal because it requires x[n+1].

(e) Explain the trade-off between stronger smoothing and temporal localization.

Solution. Longer averages smooth more, but smear rapid changes over a longer time and introduce greater effective delay.

14. Equivalent FIR descriptions and implementations

An FIR system has impulse response $h[n] = \delta[n] - 2\delta[n - 1] + 2\delta[n - 2] - \delta[n - 3]$

(a) Write the coefficient list b[0],...,b[3].

Answer: $b[0] = 1,\ \ b[1] = -2,\ \ b[2] = 2,\ \ b[3] = -1$

(b) Write the difference equation.

Answer: $y[n] = x[n] - 2x[n-1] + 2x[n-2] - x[n-3]$

(c) Draw a direct-form implementation.

Solution. Use three unit delays and four multiplier branches with coefficients 1, −2, 2, and −1, summed at the output.

(d) Explain why any alternative block diagram that produces exactly the same difference equation is an equivalent realization of the filter.

Solution. Because it has the same input-output relationship for every input.

(e) For an input sequence of length 10, state the maximum length of the complete convolution output.

Answer: $10 + 4 - 1 = 13 \text{ samples}$

15. Integrated computational investigation

Use Python with NumPy and Matplotlib to investigate FIR filtering and convolution.

(a) Create the finite input x[n]={2,4,6,4,2} and implement causal 3-point and 7-point moving-average filters. Plot the input and outputs.

Answer: $y_3 = \{2/3, 2, 4, 14/3, 4, 2, 2/3\}$

Answer: $y_7 = \{2/7, 6/7, 12/7, 16/7, 18/7, 18/7, 18/7, 16/7, 12/7, 6/7, 2/7\}$

(b) For h[n]={3,−1,2,1}, compute the convolution with x[n] both manually and with numpy.convolve, and verify that the results match.

Answer: $x * h = \{6, 10, 18, 16, 18, 12, 8, 2\}$

(c) Generate the unit impulse and verify numerically that filtering it returns the coefficient sequence h[n].

Answer: $\delta[n] * h[n] = h[n] = \{3, -1, 2, 1\}$

(d) Implement the first-difference filter and apply it to a piecewise-constant sequence. Plot the detected transitions.

Solution. The output is zero on constant sections and nonzero only where the input changes; the output value equals the signed jump size.

(e) Convolve two FIR impulse responses in both orders and verify commutativity. Then compare a cascade with the single equivalent FIR filter.

Answer: $h_1 * h_2 = h_2 * h_1$

Solution. The cascade output is identical to filtering once with $h_{eq}=h_1*h_2$.

(f) Summarize how the numerical experiments illustrate impulse response, convolution, causality, smoothing, and the LTI properties.

Solution. Impulse input reveals h[n]; convolution predicts all outputs; causal FIRs use current/past samples; longer averages smooth more; convolution commutativity/associativity explain equivalent cascades.